

CELLULAR ADAPTATION, INJURY, DEATH

Hypertrophy – increase in the SIZE of cells

- Physiologic –“normal”: increased functional demands, stimulation by growth factors, hormones (examples – uterine growth during pregnancy, skeletal muscle in response to increased demand – e.g body building)
- Pathologic- “abnormal” – examples – heart in hypertension, aortic stenosis
- Note: Certain types of cells in tissues , mainly skeletal muscle, cardiac muscle, and nerve tissue, do not proliferate because they are considered permanent tissue. Therefore they can only get bigger via hypertrophy

Hyperplasia – increase in the NUMBER of cells

- Physiologic – hormonal, growth factors (breast at puberty/pregnancy)
- Pathological – e.g inappropriate or excessive hormonal (endometrial hyperplasia) or growth factors acting on target cells

Hyperplasia and hypertrophy can occur together in some tissues/organs but usually not in skeletal and cardiac muscle, and nerve tissue.

METAPLASIA – CHANGE FROM ONE CELL TYPE TO ANOTHER TO ADAPT TO ENVIRONMENT, ETC. THIS IS A REVERSIBLE CELLULAR ADAPTATION. SEE EXAMPLES IN THE POWER POINT SLIDES. E.G., BARRETT’S ESOPHAGUS

Atrophy

- Physiologic - normal development, uterus post partum
- Pathologic – disuse (decreased workload), denervation, decreased blood supply, decreased nutrition, decreased hormone stimulation, pressure

Cell injury:

- Damage to mitochondria causes decreased oxidative phosphorylation and therefore decreased ATP
- Reduced Na/K pump leads to cell swelling (hydropic change, vacuolar degeneration) aka edema – reversible because Na enters the cell and water follows Na.
- Swelling (edema) of cells is reversible injury and happens when the Na/K pump does not function properly because of the loss of ATP. This pump is a plasma membrane pump.

- **After 20 – 30 minutes, changes become irreversible:**
 - Increase in anaerobic glycolysis – decreases glycogen, increases lactic acid, decreases pH (chromatin clumping)
 - Decrease in protein synthesis
 - **Failure of CA pump----Ca enters cell causing membrane and nuclear damage via activation of enzymes such as phospholipases, proteases, endonuclease, and ATPases (irreversible)**
 - Denaturation of intracellular proteins, enzymatic digestion of cell, contents leak out and elicit inflammation

Coagulative necrosis: Architecture (cell outlines) preserved, nucleus disappears (breaks up, disintegrates- pyknosis, karyohexis, karyolysis) – loss of nucleus with outline of cell “ghost” remaining.

An example includes a myocardial infarct in which coagulative necrosis would show nuclear loss.

Local area of coagulative necrosis – infarct

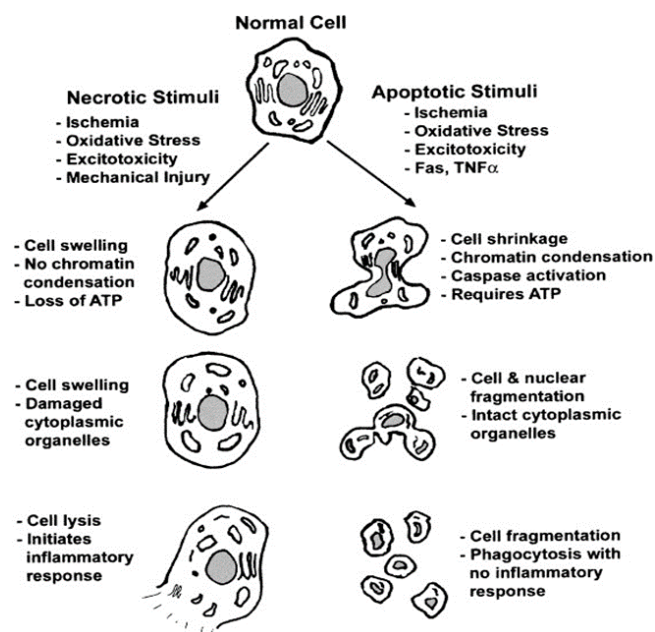
Liquefactive necrosis: Digestion of dead cells resulting in liquid, viscous mass, examples include brain infarcts (strokes), abscesses

Fat necrosis – breast, pancreatitis

Granulomatous necrosis – TB

Fibrinoid necrosis - vasculitis

Necrosis versus apoptosis (programmed cell death). Necrosis is always pathological. Apoptosis could be physiologic or pathologic and requires activation by caspases. This figure should be in power point.



Examples of pigments that may accumulate in cells include:

Anthraxis – especially in lungs in heavy smokers, polluted areas, coal, dust

Hemosiderin – old blood

Lipofuscin – “wear and tear” pigment associated with aging cells

INFLAMMATION/HEALING AND REPAIR

Timing of inflammation:

Neutrophils (most predominant around 24 hours) , macrophages (most predominant around 48 – 72 hours) , granulation tissue (new blood vessels aka angiogenesis, macrophages, and fibroblasts to produce collagen). Granulation tissue is usually present by one week (5 – 7 days). The fibroblasts mainly lay down collagen and ECM and after about 1 – 2 months fibrous scar is present (especially in permanent tissue and large skin wounds)

Neutrophils first roll via selectins, then become activated and adhere to endothelium via integrins, then transmigrate across the endothelium (pierce the basement membrane) and migrate toward chemoattractants emanating from the source of injury. Different molecules play predominant roles in different steps of this process: selectins, in rolling; chemokines (usually displayed bound to proteoglycans), in activating the neutrophils to increase avidity of integrins; integrins, in firm adhesion.

Note: Neutrophils are not only the “first responders” in acute inflammation but are also the predominant inflammatory cell in acute infections especially acute bacterial infections.

Granulomas:

Chronic inflammation - attempt to contain offending agent difficult to eradicate, e.g. Mycobacteria tuberculosis (TB)

- Macrophages process and present antigen to helper T cells.
- This interaction produces cytokines like IL-2 which activate other T cells (TH1 subtype) perpetuating response by causing secretion of IFN gamma
- IFN gamma leads to macrophage activation, converting macrophages to epithelioid histiocytes and giant cells
- Necrotizing (caseating) v. non-necrotizing (non-caseating)

From Summary of Inflammation

- Vascular changes of inflammation (vasodilation, vascular permeability, edema, stasis, margination)
- Major mediators involved in inflammation (histamine, NO, bradykinin, prostaglandins, leukotrienes, cytokines, complement)
- Cellular response in inflammation (margination, rolling, adhesion, transmigration, chemotaxis) including roles of adhesion selectins and integrins
- Phagocytosis – Recognition (Opsonization), Engulfment, Killing
- Termination of acute inflammation

- Types of acute inflammation – serous, fibrinous, suppurative
- Chronic inflammation including cell types – macrophages, lymphocytes, eosinophils, granulomas
- Acute v. Chronic inflammation (in general, think neutrophils for acute and mononuclear cells such as lymphocytes and macrophages for chronic.
 - Eosinophils are also considered chronic inflammatory cells: allergies, asthma, parasites.

Note:

CD markers are important and are detected usually via flow cytometry. For now remember that low numbers under 10 are usually T lymphocytes and CD10, CD 19, CD20, etc. are usually B lymphocytes. However there are many, many CDs and there is no need to memorize them all now.

The usual peripheral blood (NOT bone marrow) differential in a healthy adult would be about 65 – 75% neutrophils, 30 – 40 % lymphocytes, monocytes, eosinophils, 1- 2 % basophils. These are just ranges and labs may differ somewhat in normal values.

The lymphocytes which make up the 30 – 40% of the peripheral blood are mostly T lymphocytes (low number CD markers). However, the only way to tell the difference between T and B lymphocytes is using CD markers. A peripheral blood smear cannot distinguish the different types of lymphocytes. Most of the lymphocytes in the blood are T lymphocytes but you can not distinguish them by just looking at them with a light microscope.

Review of Healing: (Skin Example)

- 24 to 48 hours - epithelial cells move from wound edge along cut margins of dermis, deposit basement membrane components; produce thin, continuous epithelial layer; macrophages predominate (remember neutrophils arrive first and are prominent by 24 hours)
- Fibroblasts, vascular endothelial cells proliferate to form granulation tissue by 5 to 7 days, fills wound, neovascularization maximal. Granulation tissue is mainly made up of new vessels, macrophages and fibroblasts. The fibroblasts begin to deposit collagen which will eventually form scar tissue.
- Wbc infiltrate, edema, increased vascularity disappear by second week
- TGF-beta is most important fibrogenic agent: causes fibroblast migration and proliferation, increased synthesis of collagen and fibronectin, and decreased ECM degradation by MMPs
- Increased collagen accumulation and regression of vascular channels – original granulation tissue scaffold converted to pale, avascular scar made up of fibroblasts, dense collagen, elastic tissue, other ECM components
- By end of first/second month – scar is composed of acellular connective tissue without inflammation, covered by intact epithelium

Some important mediators:

- PDGF causes migration and proliferation of fibroblasts, smooth muscle cells, monocytes to inflammatory areas and healing skin wounds

- VEGF important in new blood vessel formation (angiogenesis), promotes angiogenesis in chronic inflammation, wound healing, and tumors
- TGF-beta is potent fibrogenic agent - stimulates fibroblast chemotaxis, enhances production of collagen, fibronectin, and proteoglycans; inhibits collagen degradation, anti-inflammatory

Wound tensile strength plateaus at about 70-80% around 3 months – never really returns to full strength.

Primary v secondary intention healing – examples of healing in skin wounds. Primary intention occurs with surgical or clean linear wounds in which the skin edges are apposed to each other and there is not a lot of tissue lost on the surface.

Deterrents to healing may include debris within wound, obesity, vomiting, dehiscence, keloid formation, etc. If the wound has debris in it such as foreign objects, dirt, glass, rocks, etc. it must be debrided.

NEOPLASIA

Nomenclature is important. Adenoma v. Adenocarcinoma (this applies to glandular epithelium), etc.

Squamous cell carcinoma, which is not glandular, if well differentiated will produce keratin (should not be present in a pure adenocarcinoma).

A tumor may be benign and epithelial – for example adenoma

A tumor may be benign and mesenchymal – for example LIPOMA. Mesenchymal refers to connective tissue and other elements such as adipose tissue, fibrous tissue, muscle, bone, etc.

A tumor may be malignant and epithelial (most are) – for example, ADENOCARCINOMA (colon, lung, etc.)

A tumor may be malignant and mesenchymal (rare) – for example, liposarcoma, fibrosarcoma, osteosarcoma, leiomyosarcoma, rhabdomyosarcoma.

Please note: While they are both malignant tumors, a carcinoma is derived from epithelial tissue and a sarcoma is derived from mesenchymal tissue. Prognosis and treatments are different and carcinomas are more common in the general population.

What is the definitive criterium for malignancy? THE ABILITY TO METASTASIZE.

Hallmarks of Malignancy

1. Proliferate without external stimuli (growth signals), usually because of oncogene activation – self-sufficiency
2. Insensitive to growth-inhibitory signals

3. Undergo metabolic switch to aerobic glycolysis (Warburg effect) which enables synthesis of macromolecules needed for fast cell growth. See below for further explanation
4. Evade apoptosis (p53 inactivation or activation of anti-apoptotic genes)- tumors are resistant to programmed cell death
5. Have limitless proliferative capacity = immortality (telomerase)**
6. Sustained angiogenesis
7. Ability to invade and metastasize
8. Evade host immune response

****The Enzyme telomerase is very important for one of the hallmarks of cancer – it gives cells unlimited replicative potential by not allowing the telomeres to continue to shorten as they would in a normal cell which has a limited lifetime.**

- **Telomeres - sequence of repetitive bases at end of linear chromosomes**
 - prevent chromosomes from attaching
 - determine limited number of cell replications
 - **TELOMERASE**
 - enzyme present in 90% of human cancers; most somatic cells do not express telomerase
 - changes telomeres so they have unlimited replicative potential
- Terms to associate with malignancy (metastasis is definitive)**

- **Anaplasia – lack of differentiation, “to form backward”**
- **Well to moderately to poorly differentiated refers to how neoplasms appears when seen through microscopic – this is associated with GRADE of tumor.**
- **Malignant – ability to metastasize**
 - Pleomorphism – variation in size and shape of cells
 - Abnormal nuclear morphology
- **Hyperchromasia**
- **Increased nuclear/cytoplasm ratio – may approach 1:1 ratio**
- **Increased mitotic rate (proliferative activity); bizarre mitotic figures more reliable**
- **Loss of polarity – tumor cells grow in disorganized fashion**
- **Necrosis – outgrowth of blood supply, especially in large tumors**

Words to associate with malignancy (although metastases is the definitive criteria): Anaplasia, pleomorphic, invasion/infiltration, metastases, necrosis, high mitotic rate (INCREASE IN MITOSES), hyperchromasia, increased nuclear to cytoplasm ratio. Note: Dysplasia can also have hyperchromasia, increased N/C ratio, higher mitotic rate but to a lesser degree than neoplasia. However dysplasia is NOT associated with invasion/infiltration.

Masses which are poorly circumscribed are also more likely to be malignant.

Differentiation: Helps determine GRADE of tumor – extent that the neoplastic cells resemble corresponding normal cells, morphologically and functionally

Grade:

- **Based on cellular appearance and mitotic activity microscopically**

- Based on degree of differentiation; therefore, low grade = well-differentiated but high grade would imply poorly differentiated (does not resemble original tissue from which it developed)
- idea that biological behaviour and differentiation are related, i.e., poorly differentiated tumors behave more aggressively. Correlation is not perfect.

Stage:

- Degree of localization/spread based on size, local and regional lymph node spread, and distant metastases
 - determined clinically by surgery and/or radiology; more clinically relevant
- TNM Tumor, Nodes, Metastases – Staging system
- T0 = in situ

Note: Spread/metastasis to lymph nodes (regional or distant) will increase the stage of the cancer.

Tumor markers in the serum may aid more with follow-up than detection. Some are not specific. However please be familiar with alpha-fetoprotein (AFP) in hepatocellular carcinoma, CEA, CA-19-9, PSA, etc.

Certain viruses are oncogenic (induce cancer development) – examples include HPV (human papillomaa virus) in cervical squamous cell carcinoma, EBV, etc.

Helicobacter pylori is a bacteria which if present chronically in the stomach may induce a low grade lymphoma.

Regulatory genes involved in cancer: 4 major categories

- Proto-oncogenes - involved in normal cell growth and repair (Abl, HER, Myc, Ras) – growth promoting
- DNA repair genes
- Tumor suppressor genes - protect against unregulated cell growth (APC, BRCA) – growth inhibiting
- Antiapoptosis genes: overexpression prevents apoptosis (Bcl-2); apoptosis genes mutations prevent programmed cell death, i.e promotes proliferation

Examples of important genes in cancer: for now these 3 are especially important: RAS, MYC, and p53

RAS Oncogene, example – Signal transducer

Most common type of abnormality involving proto-oncogenes in humans

- HRAS, KRAS, NRAS
- Point mutation of RAS genes - single most common abnormality of proto-oncogenes in human tumors
- 15-20% of all human cancers express mutated RAS
- Usually carcinomas, esp. colon and pancreas, are associated with KRAS, bladder tumors with HRAS, and hematopoietic tumors with NRAS

- Normal RAS proteins flip between an excited signal-transmitting state bound to GTP and quiescent state bound to GDP; stimulation of receptor tyrosine kinase by growth factors leads to exchange of GDP for GTP to make active RAS
- Mutated RAS protein is permanently activated (stimulates downstream regulators of proliferation) because of inability to hydrolyze GTP, causing continuous stimulation of cells without external trigger

MYC Oncogene – Nuclear regulator

Activation linked to proliferation; activates expression of many genes involved in cell growth – encodes for transcription factors and also involved with apoptosis

- Many ways MYC can be deregulated in cancer cells
- Translocation of the gene causes dysregulation of Myc expression in Burkitt lymphoma
- Amplified in breast, colon, lung and other cancers which results in overexpression
- Sometimes MYC upregulates expression of telomerase (contributes to endless replicative capacity) – see later slide
- N-Myc and L-Myc amplified in neuroblastoma and small cell cancer of lung, respectively

p53 – Tumor suppressor gene

Tumor suppressor gene mutated in majority of cancers; on chromosome 17

- NORMALLY regulates cell cycle progression, DNA repair, senescence, and apoptosis
- NORMALLY prevents neoplastic transformation via
 - Temporary cell cycle arrest (quiescence)
 - Induction of permanent cell cycle arrest (senescence)
 - Triggering programmed cell death (apoptosis)
- NORMALLY Acts mainly via cell cycle inhibitor p21 to cause cell cycle arrest
- NORMALLY Causes apoptosis by inducing transcription of genes such as BAX
- Required for G1/S checkpoint, main component of G2/M checkpoint
- p53 aka “molecular policeman”: prevents propagation of genetically damaged cells
 - Homozygous loss occurs in almost all types of cancer e.g. lung, colon, breast
 - Over 50% of tumors have mutations of p53 – biallelic loss-of-function mutations
 - Most sporadic cancers require “two hits”, that is, both alleles need to be mutated.
- Li-Fraumeni syndrome – VERY IMPORTANT
 - patients inherit one mutant p53 allele; THEREFORE, only require one “hit” to lose p53 functions stated above.
 - 25-fold greater chance of developing a malignant tumor by age 50 than general population
 - sarcomas, breast cancer, leukemia, brain tumors, adrenal cortex carcinomas, and others

Paraneoplastic syndromes: May represent early manifestation of occult neoplasm

- 10% of persons with malignant disease
- May be earliest manifestation of occult neoplasm
- Endocrinopathies
- Nerve and muscle syndromes – e.g., myasthenia gravis (thymoma)
- Dermatological disorders
- Vascular and hematological changes

- Others – bone, joint, soft tissue changes, nephrotic syndrome

Warburg effect Note:

The Scholars Rx brick, which mentions the Warburg effect in neoplasia, is not very clear. I have annotated it and updated it, and also copied and pasted the direct text from Robbins below which may be more helpful: Please note that even Robbins can be wrong, or in this case, have a typo. See below- fermentation via the glycolytic pathway converts glucose to lactate NOT lactose. Thank you Daniel Chan Class of 2026 for picking that up!

“Even in the presence of ample oxygen, cancer cells demonstrate a distinctive form of cellular metabolism characterized by high levels of glucose uptake and increased conversion of glucose to lactose (fermentation) via the glycolytic pathway. This phenomenon, called the Warburg effect and also known as aerobic glycolysis, has been recognized for many years... Clinically the “glucose hunger” of tumors is used to visualize them via positron emission tomography (PET) scanning, in which patients are injected with 18 F-fluorodeoxyglucose, a nonmetabolizable derivative of glucose that is preferentially taken up into tumor cells (as well as normal, actively dividing tissues such as the bone marrow). Most tumors are PET-positive, and rapidly growing ones are markedly so.

...why is it advantageous for a cancer cell to rely on seemingly inefficient glycolysis (which generates two molecules of ATP per molecule of glucose) instead of oxidative phosphorylation (which generates 36 molecules of ATP per molecule of glucose)? While pondering this question, it is important to recognize that rapidly growing normal cells, such as in embryonic tissues, also rely on aerobic fermentation. Thus, “Warburg metabolism” is not cancer-specific, but instead is a general property of growing cells that is exploited by cancer cells.

The answer to this riddle is surprisingly simple: aerobic glycolysis provides rapidly dividing tumor cells with metabolic intermediates that are needed for the synthesis of cellular components, whereas mitochondrial oxidative phosphorylation does not. The reason growing cells rely on aerobic glycolysis becomes readily apparent when one considers that a growing cell has a strict biosynthetic requirement; it must duplicate all of its cellular components—DNA, RNA, proteins, lipid, and organelles—before it can divide and produce two daughter cells. Recall that the net effect of oxidative phosphorylation is to take a molecule of glucose, $C_6H_{12}O_6$, and combine it with six molecules of O_2 to produce six molecules of H_2O and six molecules of CO_2 , which are lost through respiration. Thus, while “pure” oxidative phosphorylation yields abundant ATP, it fails to produce any carbon moieties that can be used to build cellular components that are needed for growth (proteins, lipids, and nucleic acids). Even cells that are not actively growing must shunt some metabolic intermediates away from oxidative phosphorylation to synthesize macromolecules that are needed for cellular maintenance.

By contrast, in actively growing cells, only a small fraction of the cellular glucose is shunted through the oxidative phosphorylation pathway, such that on average each molecule of glucose that is metabolized produces approximately four molecules of ATP (instead of the two molecules that would be produced by “pure” glycolysis). Presumably, this balance in glucose utilization (heavily biased toward aerobic fermentation, with a bit of oxidative phosphorylation) hits a metabolic “sweet spot” that is optimal for growth. It follows that growing cells do rely on mitochondrial metabolism. However, the main metabolic function of mitochondria in growing cells is not to generate ATP, but rather to carry out reactions that

generate intermediates that can be diverted for use as precursors in the synthesis of cellular building blocks..”

THE NEW EDITION OF ROBBINS CAME OUT THIS PAST JUNE 2025, AND I HAVE MADE SURE THAT EVERYTHING IN THE POWER POINT, ETC. HAS BEEN UPDATED IF NEEDED, EVEN THOUGH THE SLIDE MAY STILL HAVE THE 2020 REFERENCE. I JUST DID NOT CHANGE THE YEAR YET on the slide.

The questions below are taken from Robbins and Cotran Review of Pathology 3rd edition. These tend to be more on the difficult side, so again, it is not about getting them all correct right now. Some minor edits have been made.

Answers are at the end of each section.

Please read the explanations which may provide useful information.

There are 6 questions pertaining to Inflammation and 5 questions related to Healing and Repair.

Acute and Chronic Inflammation

1. A 63-year-old man develops worsening congestive heart failure 2 weeks after an acute myocardial infarction. An echocardiogram shows a markedly decreased ejection fraction. He dies 1 day later. At autopsy, a section of the infarct shows that the necrotic myocardium has largely been replaced by capillaries, fibroblasts, and collagen. Various inflammatory cells are present. Which of the following inflammatory cell types in this lesion plays the most important role in the healing process?

- (A) Macrophages
- (B) Plasma cells
- (C) Neutrophils
- (D) Eosinophils
- (E) Epithelioid cells

2. A 36-year-old man has had midepigastria abdominal pain for the past 3 months. An upper gastrointestinal endoscopy shows a 2-cm, sharply demarcated, shallow ulceration of the gastric antrum. A biopsy specimen of the ulcer base shows angiogenesis, fibrosis, and mononuclear cell infiltrates with lymphocytes, macrophages, and plasma cells. Which of the following terms best describes this pathologic process?

- (A) Acute inflammation
- (B) Serous inflammation
- (C) Granulomatous inflammation
- (D) Fibrinous inflammation
- (E) Granulation tissue formation

3. A 5-year-old child reaches up to the stove and touches a pot of boiling soup. Within several hours, there is marked erythema of the skin of the fingers on the child's right hand, and small blisters appear on the finger pads. Which of the following terms best describes this process?

- ☐ (A) Fibrinous inflammation
- ☐ (B) Purulent inflammation
- ☐ (C) Serous inflammation
- ☐ (D) Ulceration
- ☐ (E) Granulomatous inflammation

4. A 6-year-old child has a history of recurrent infections with pyogenic bacteria, including *Staphylococcus aureus* and *Streptococcus pneumoniae*. The infections are accompanied by a neutrophilic leukocytosis. Microscopic examination of a biopsy specimen obtained from an area of soft tissue necrosis shows microbial organisms, but very few neutrophils. An analysis of neutrophil function shows a defect in rolling. This child's increased susceptibility to infection is most likely caused by a defect in which of the following molecules?

- ☐ (A) Selectins
- ☐ (B) Integrins
- ☐ (C) Leukotriene B₄
- ☐ (D) Complement C3b
- ☐ (E) NADPH oxidase

5. In an experiment, neutrophils collected from peripheral blood are analyzed for a "burst" of oxygen consumption. This respiratory burst is an essential step for which of the following events in an acute inflammatory response?

- ☐ (A) Increased production in bone marrow
- ☐ (B) Attachment to endothelial cells
- ☐ (C) Opsonization of bacteria
- ☐ (D) Phagocytosis of bacteria
- ☐ (E) Generation of microbicidal activity

6. A 70-year-old woman has had worsening shortness of breath for the past week. On physical examination, her temperature is 102 degrees F. On percussion, there is

dullness over the left lung fields. Thoracentesis (removal of fluid from the thoracic cavity) yields 800 mL of cloudy yellow fluid from the left pleural cavity. Analysis of the fluid reveals a WBC count of 2500/mm³ with 98% neutrophils and 2% lymphocytes. A Gram stain of the fluid shows the presence of bacteria. Which of the following terms best describes the process occurring in the left pleural cavity?

- ☐ (A) Serous effusion
- ☐ (B) Chronic inflammation
- ☐ (C) Abscess
- ☐ (D) Fibrinous inflammation
- ☐ (E) Purulent exudate

ANSWERS

1. **(A)** Macrophages, present in such lesions, play a prominent role in the healing process. Activated macrophages can secrete various cytokines that promote angiogenesis and fibrosis, including platelet-derived growth factor, fibroblast growth factor, interleukin-1, and tumor necrosis factor. Plasma cells can secrete immunoglobulins and are not instrumental to healing of an area of tissue injury. Neutrophils are most numerous within the initial 48 hours after infarction, but are not numerous after the first week. They are important in the acute inflammatory response. Eosinophils are most prominent in allergic inflammations and in parasitic infections. Epithelioid cells, which are aggregations of activated macrophages, are typically seen with granulomatous inflammation. The healing of acute inflammatory processes does not involve granulomatous inflammation.
2. **(E)** Granulation tissue formation is an important part of the healing process. As the acute inflammatory response wanes, granulation tissue forms to start the regeneration/repair process. Granulation tissue is predominantly made of up new blood vessels, macrophages, and fibroblasts.
3. **(C)** Serous inflammation is the mildest form of acute inflammation. A blister is a good example of serous inflammation. It is associated primarily with exudation of fluid into the subcorneal or subepidermal space. Because the injury is mild, the fluid is relatively protein-poor. A protein-rich exudate results in fibrin accumulation. Acute inflammatory cells, mainly neutrophils, exuded into a body cavity or space form a purulent (suppurative) exudate, typically associated with liquefactive necrosis. Loss of the epithelium leads to ulceration. Granulomatous inflammation is characterized by collections of transformed macrophages called epithelioid cells.
4. **(A)** The patient has a defect in leukocyte rolling, the first step in transmigration of neutrophils from the vasculature to the tissues. Rolling depends on interaction between selectins (P-selectin and E-selectin on endothelial cells, and L-selectin on

neutrophils) and their sialylated ligand molecules (e.g., sialylated Lewis X). Integrins are involved in the next step of transmigration, during which there is firm adhesion between neutrophils and endothelial cells. Leukotriene B₄ is a chemotactic agent, complement C3b facilitates phagocytosis, and NADPH oxidase is involved in microbicidal activity.

5. **(E)** The respiratory, or oxidative, burst generates reactive oxygen species (i.e., superoxide anion) that are important in destruction of engulfed bacteria. Myelopoiesis does not depend on generation of superoxide. Endothelial attachment of neutrophils is aided by adhesion molecules on the endothelium and the neutrophil surface. These molecules include selectins and integrins. Bacteria are opsonized by complement C3b and IgG, allowing the bacteria to be more readily phagocytosed.
6. **(E)** Bacterial infections often evoke an acute inflammatory response dominated by neutrophils. The extravasated neutrophils attempt to phagocytose and kill the bacteria. In the process, some neutrophils die, and the release of their lysosomal enzymes can cause liquefactive necrosis of the tissue. This liquefied tissue debris and the live and dead neutrophils comprise pus, or purulent exudate. Such an exudate is typical of bacterial infections that involve body cavities. Another term for purulent exudate in the pleural space is *empyema*. An abscess is a localized collection of neutrophils within tissues. Chronic inflammation occurs when there is a preponderance of mononuclear cells, such as lymphocytes, macrophages, and plasma cells, in a process that has gone on for more than a few days—more likely weeks or months— or that accompanies repeated bouts of acute inflammation. Edema refers to increased fluid collection within tissues, leading to tissue swelling. In fibrinous inflammation, exudation of blood proteins (including fibrinogen, which polymerizes to fibrin) gives a grossly shaggy appearance to surfaces overlying the inflammation. A serous effusion is a watery-appearing transudate that resembles an ultrafiltrate of blood plasma, with a low cell and protein content.

Healing/Repair

1. In an experiment, surgical incisions are made in a study group of laboratory rats. Observations about the wounds are recorded over a 2-week period using various chemical mediators. Which of the following steps in the inflammatory-repair response is most likely affected by neutralization of transforming growth factor β (TGF- β)?

- ☐ (A) Leukocyte extravasation
- ☐ (B) Increase in vascular permeability
- ☐ (C) Production of collagen
- ☐ (D) Chemotaxis of lymphocytes
- ☐ (E) Migration of epithelial cells

2. A cesarean section is performed on a 20-year-old woman to deliver a term infant, and the lower abdominal incision is sutured. The sutures are removed 1 week later. Which of the following statements best describes the wound site at the time of suture removal?

- ☐ (A) Granulation tissue is still present
- ☐ (B) Collagen degradation exceeds synthesis
- ☐ (C) Wound strength is 80% of normal tissue
- ☐ (D) Type IV collagen predominates
- ☐ (E) There is a predominance of neutrophils

3. A 40-year-old man underwent laparotomy for a perforated sigmoid colon diverticulum. A wound infection complicated the postoperative course, and surgical wound dehiscence occurred. Primary closure was no longer possible, and the wound "granulated in." Six weeks later, the wound is only 10% of its original size. Which of the following processes best accounts for the observed decrease in wound size over the past 6 weeks?

- ☐ (A) Increase in synthesis of collagen
- ☐ (B) Myofibroblast contraction
- ☐ (C) Inhibition of metalloproteinases
- ☐ (D) Resolution of subcutaneous edema
- ☐ (E) Elaboration of adhesive glycoproteins

4. An 18-year-old man lacerated his left hand and required sutures. The sutures were removed 1 week later. Wound healing continued, but the site became disfigured by a prominent raised, nodular scar that developed over the next 2 months. Which of the following terms best describes the process that occurred during this 2-month period?

- ☐ (A) Organization
- ☐ (B) Dehiscence
- ☐ (C) Resolution
- ☐ (D) Keloid formation
- ☐ (E) Secondary union

5. A 24-year-old man with acute appendicitis undergoes surgical removal of the inflamed appendix. The incision site is sutured. Approximately how long after the surgery would scar tissue be evident?

- ☐ (A) 1 day
- ☐ (B) 2 to 3 days
- ☐ (C) 4 to 5 days
- ☐ (D) 2 weeks
- ☐ (E) 1 to 2 month

ANSWERS

1. **(C)** TGF- β stimulates many steps in fibrogenesis, including fibroblast chemotaxis and production of collagen by fibroblasts, while inhibiting degradation of collagen. All of the other steps listed are unaffected by TGF- β .
2. **(A)** At 1 week, wound healing is incomplete, and granulation tissue is still present. More collagen is synthesized in the following weeks. Wound strength peaks at about 80% by 3 months. Type IV collagen is found in basement membranes.
3. **(B)** Wound contraction is a characteristic feature of healing by second intention that occurs in larger wounds. Collagen synthesis helps fill the defect, but does not contract it. The inhibition of metalloproteinases leads to decreased degradation of collagen and impaired connective tissue remodeling in wound repair. Edema diminishes over time, but this does not result in much contraction. Adhesive glycoproteins such as fibronectin help to maintain a cellular scaffolding for growth and repair, but they do not contract.
4. **(D)** The healing process sometimes results in an exuberant production of collagen, giving rise to a keloid. This tendency may run in families. Organization occurs as granulation tissue is replaced by fibrous tissue. Dehiscence occurs when a wound pulls apart. If normal tissue architecture is restored, resolution of inflammation has occurred. Secondary union describes the process by which large wounds fill in and contract.
5. **(E)** Dense collagen (scar tissue) with some remaining dilated blood vessels, typical of the final phase of wound healing, is extensive by the end of the first month. On day 1, the wound is filled only with fibrin and inflammatory cells. Macrophages and granulation tissue are seen 2 to 3 days postoperatively. Neovascularization is most prominent by days 4 and 5. By week 2, collagen is prominent, and fewer vessels and inflammatory cells are seen.